BLUESTOCK FINTECH

CAPSTONE PROJECT

CAPSTONE PROJECT I - MUTUAL FUND ANALYTICS

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1. EXECUTIVE SUMMARY

The Mutual Fund Analytics Platform was developed as part of the Bluestock Fintech Capstone Internship Project. The objective of this project was to design and implement an end-to-end data analytics solution capable of processing, analyzing, and visualizing mutual fund industry data from multiple sources.

The project involved building a complete data pipeline beginning with data ingestion, transformation, validation, storage, advanced analytics, and dashboard reporting. Ten datasets containing information related to mutual fund schemes, NAV history, AUM growth, SIP inflows, investor transactions, portfolio holdings, benchmark indices, and performance metrics were integrated into a unified analytical platform.

Python, Pandas, NumPy, SQLite, SQLAlchemy, Jupyter Notebook, and Power BI were used to develop the solution. Advanced financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and Sector Concentration Index were calculated to evaluate mutual fund performance.

The final deliverables include an automated ETL pipeline, a relational database, exploratory data analysis notebooks, advanced analytics modules, a recommendation engine, and a multi-page Power BI dashboard that enables users to explore industry trends, investor behavior, fund performance, and market insights.

The project demonstrates practical applications of data engineering, financial analytics, business intelligence, and data visualization in the mutual fund domain.

2. PROJECT OVERVIEW & OBJECTIVE

Bluestock Fintech is a financial technology company focused on democratising investment analytics for retail and institutional investors in India. This capstone project tasks you with building a full-stack Mutual Fund Analytics Platform that ingests publicly available data from AMFI India, transforms it through a robust ETL pipeline, stores it in a relational database, and presents insights via an interactive dashboard.

The Indian mutual fund industry has grown at a remarkable pace. As of December 2025, the industry manages over Rs. 81 lakh crore in AUM across 1,908 schemes and 26.12 crore investor folios. Monthly SIP inflows crossed an all-time high of Rs. 31,002 crore in December 2025, reflecting India's deepening equity culture.

Bluestock Fintech wants a data platform that:

- Tracks NAV movements of 40+ key mutual fund schemes from top AMCs
- Monitors AUM growth trends for the 10 largest fund houses over 4+ years
- Analyses investor behaviour patterns across geographies and demographics
- Computes risk-adjusted return metrics (Sharpe, Sortino, Alpha, Beta)
- Benchmarks fund performance against Nifty 50, Nifty 100, BSE SmallCap indices
- Provides an executive-level interactive dashboard for fund selection

#	**Objective**	**Outcome**
01	Build an ETL pipeline from raw AMFI data	Automated Python script
02	Design a normalised SQL schema for MF data	5-table star schema
03	Perform comprehensive EDA on NAV & AUM data	EDA notebook with charts
04	Compute performance & risk metrics per scheme	Metrics dashboard
05	Build an interactive BI dashboard	Power BI / Tableau file
06	Analyse investor transaction patterns	Demographic insights
07	Compare fund returns vs benchmark indices	Alpha / tracking error report
08	Document and present the entire project	PDF report + slides

3. PROBLEM STATEMENT

Despite the massive growth of India's mutual fund industry, individual investors and financial advisors often struggle to make data-driven fund selection decisions due to fragmented data, lack of unified analytics platforms, and complex financial metrics. This project solves real business problems:

P1: Data Fragmentation

NAV data, AUM data, SIP flow data and portfolio holdings are available on different sections of the AMFI website in different formats (TXT, PDF, HTML tables). No single unified database exists.

P2: Performance Comparison Gap

Investors cannot easily compare multiple funds across different AMCs on a risk-adjusted basis. Raw NAV data requires significant transformation to compute Sharpe ratio, Alpha, Beta, etc

P3: No Benchmark Tracking

Most retail investors do not know whether their fund is outperforming its benchmark index. Tracking error and information ratio require both NAV and benchmark index data.

P4: Investor Behaviour Blind Spot

AMCs and distributors have limited visibility into how demographic and geographic factors influence SIP amounts, fund preferences and redemption patterns.

P5: Slow Reporting

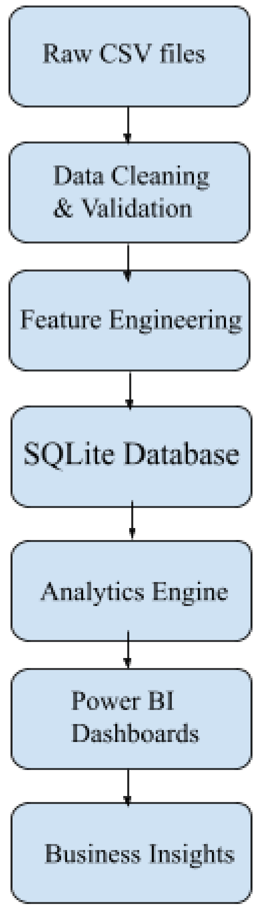
Monthly MF reports are static PDFs that take days to prepare. Stakeholders need live, self-service dashboards with drill-down capability.

4. DATASETS

Dataset	Description
01_fund_master	Fund metadata and scheme details
02_nav_history	Fund metadata and scheme details
03_aum_by_fund_house	Industry AUM information
04_monthly_sip_inflows	Monthly SIP statistics
05_category_inflows	Category-wise inflows
06_industry_folio_count	Folio statistics
07_scheme_performance	Performance metrics
08_investor_transactions	Investor transaction history
09_portfolio_holdings	Portfolio allocation details
10_benchmark_indices	Benchmark market indices

The datasets collectively provide information about mutual fund growth, investor participation, portfolio composition, and market performance from 2022 to 2025.

5. SYSTEM ARCHITECTURE



The project follows a structured analytics workflow where raw datasets are cleaned, transformed, stored in a relational database, analyzed using Python, and finally visualized using Power BI dashboards

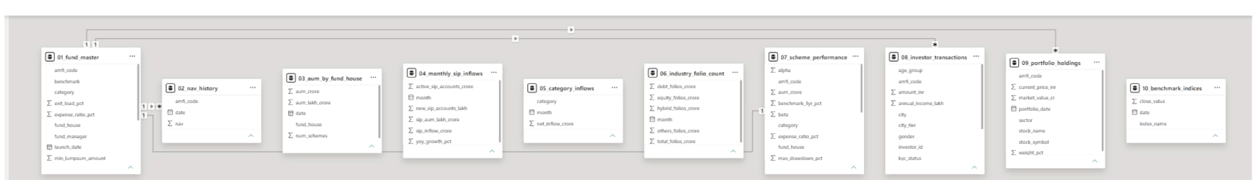


Figure 1: Data Model Relationships

6. ETL PIPELINE

Extract Phase

Data was collected from multiple CSV datasets representing different aspects of mutual fund operations.

Transform Phase

Data preprocessing activities included:

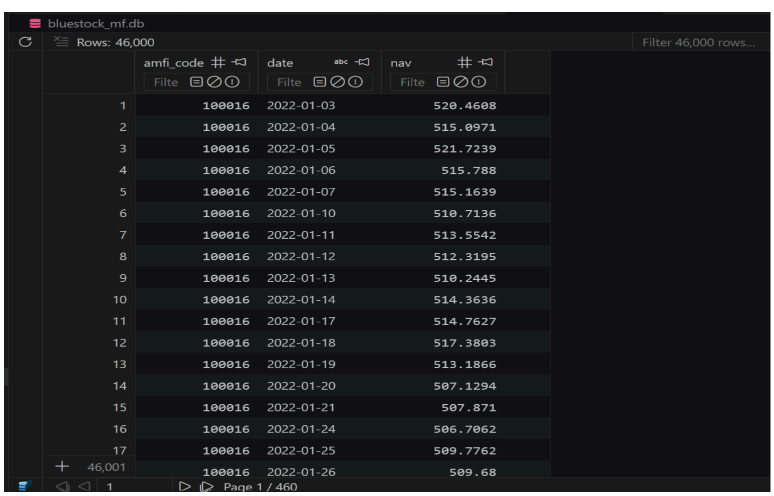
- Missing value handling
- Data type conversion
- Feature engineering
- Date formatting
- Consistency checks

Load Phase

Processed datasets were loaded into a SQLite database to support efficient querying and analytics.

Outcome

The ETL pipeline ensured high-quality, structured data suitable for advanced analytics and dashboard development.


bluestock_mf.db
Rows: 46,000

	amfi_code	date	nav
1	100016	2022-01-03	520.4608
2	100016	2022-01-04	515.0971
3	100016	2022-01-05	521.7239
4	100016	2022-01-06	515.788
5	100016	2022-01-07	515.1639
6	100016	2022-01-10	510.7136
7	100016	2022-01-11	513.5542
8	100016	2022-01-12	512.3195
9	100016	2022-01-13	510.2445
10	100016	2022-01-14	514.3636
11	100016	2022-01-17	514.7627
12	100016	2022-01-18	517.3803
13	100016	2022-01-19	513.1866
14	100016	2022-01-20	507.1294
15	100016	2022-01-21	507.871
16	100016	2022-01-24	506.7062
17	100016	2022-01-25	509.7762
	100016	2022-01-26	509.68

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Figure 2: SQLite Database Containing Processed Mutual Fund Data

7. EXPLORATORY DATA ANALYSIS (EDA)

Industry AUM Growth Trend

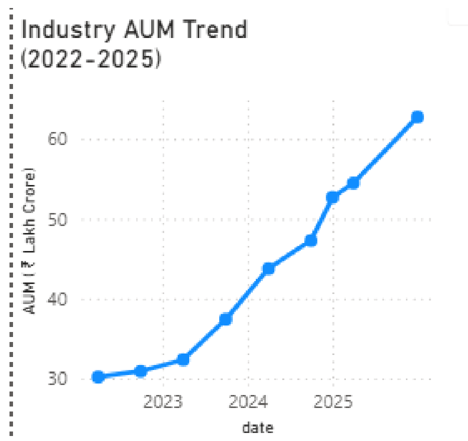


Figure 3: Industry AUM Growth Trend

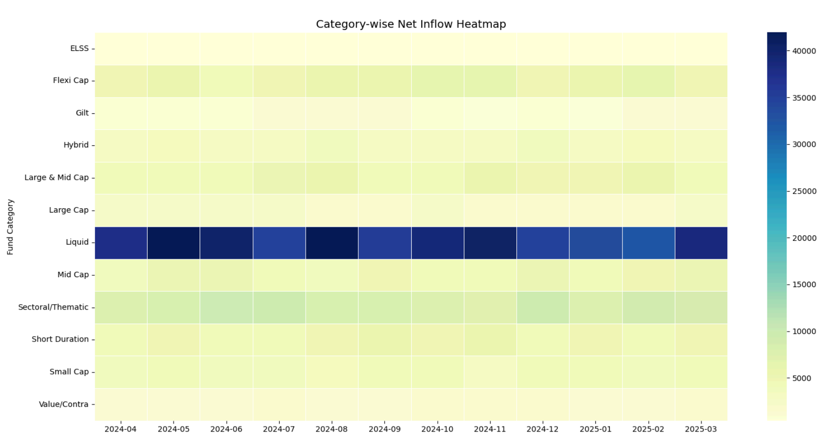
Observations

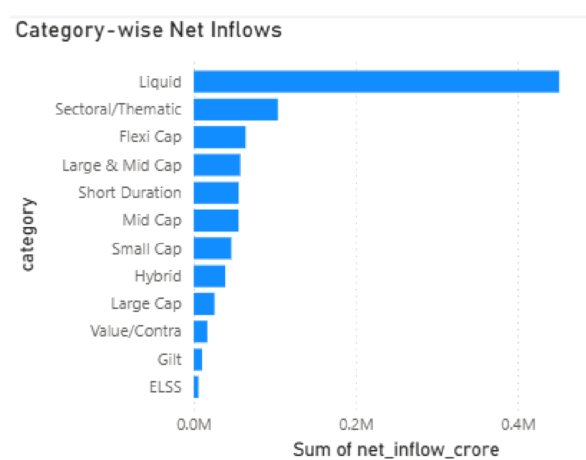
- Industry AUM increased consistently over the analysis period.
- Investor confidence in mutual funds improved significantly.
- Growth accelerated during later years.

Category-wise Net Inflow

- Liquid Funds consistently attracted the highest inflows throughout the analysis period.
- Sectoral/Thematic funds demonstrated moderate but stable investor interest.
- Large Cap, Mid Cap, and Small Cap categories maintained steady inflow patterns.
- Value/Contra and ELSS categories received comparatively lower inflows.

Figure 4 & 5: Category-wise Net Inflow Heatmap (2024–2025)





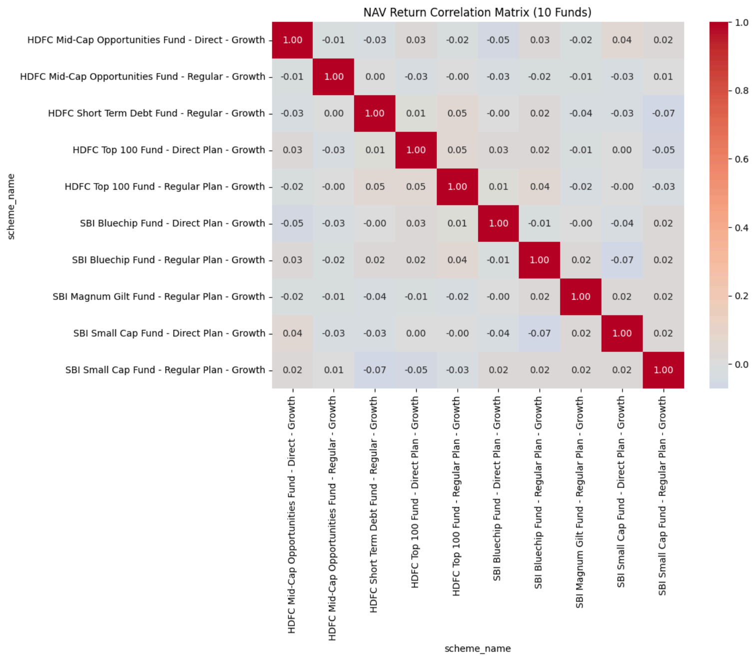


Figure 6: NAV Correlation Matrix

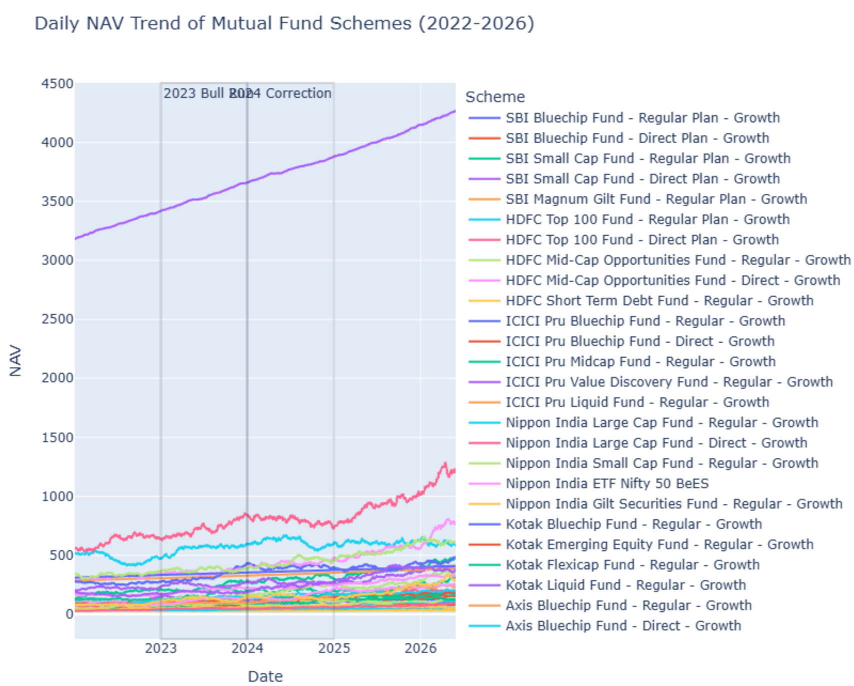


Figure 7: NAV Performance Analysis

8. FUND PERFORMANCE ANALYTICS

CAGR Analysis

CAGR was calculated to evaluate long-term growth across schemes.

Sharpe Ratio

Sharpe Ratio was used to measure risk-adjusted returns.

Alpha & Beta

Alpha measured benchmark outperformance while Beta evaluated market sensitivity.

Maximum Drawdown

Maximum Drawdown was analyzed to understand downside risk.

Fund Performance Scorecard

scheme_name	fund_house	category	Sum of return_3yr_pct	Sum of alpha	Sum of beta	Sum of sharpe_ratio	Sum of max_drawdown_pct
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Large Cap	14.81	1.62	0.96	1.06	-17.07
ICICI Pru Value Discovery Fund - Regular - Growth	ICICI Prudential MF	Value	14.76	0.55	0.92	0.98	-21.89
Mirae Asset Emerging Bluechip Fund - Regular - Growth	Mirae Asset MF	Large & Mid Cap	14.56	1.70	0.99	0.91	-33.15
ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	Large Cap	14.41	0.88	1.03	1.03	-26.59
Nippon India Large Cap Fund - Regular - Growth	Nippon India MF	Large Cap	14.00	0.86	0.88	1.00	-16.07
ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla Sun Life MF	Large Cap	13.78	1.34	1.03	0.98	-15.07
Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	ELSS	13.58	0.54	0.98	0.80	-22.62
HDFC Top 100 Fund - Direct Plan - Growth	HDFC Mutual Fund	Large Cap	13.38	1.13	0.97	0.96	-33.50
DSP Top 100 Equity Fund - Regular - Growth	DSP Mutual Fund	Large Cap	12.82	1.82	0.91	0.92	-21.70
SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	Large Cap	12.36	0.87	0.89	0.88	-21.70
Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	Large Cap	12.33	1.70	1.02	0.88	-18.14
Kotak Bluechip Fund - Regular - Growth	Kotak Mahindra MF	Large Cap	12.25	1.27	0.93	0.87	-17.86
Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund	Large Cap	12.14	1.43	0.87	0.87	-17.40
UTI Nifty 50 Index Fund - Regular - Growth	UTI Mutual Fund	Index	12.10	0.93	0.90	0.93	-24.42
Axis Bluechip Fund - Regular - Growth	Axis Mutual Fund	Large Cap	11.84	1.41	0.91	0.85	-27.54
Nippon India ETF Nifty 50 BeES	Nippon India MF	Index/ETF	11.77	1.80	1.04	0.91	-26.75
ICICI Pru Bluechip Fund - Regular - Growth	ICICI Prudential MF	Large Cap	11.54	0.66	0.96	0.82	-25.91
Total			**563.56**	**50.14**	**34.93**	**54.47**	**-768.01**

Figure 8: Fund Scorecard

9. RISK VS RETURN ANALYSIS

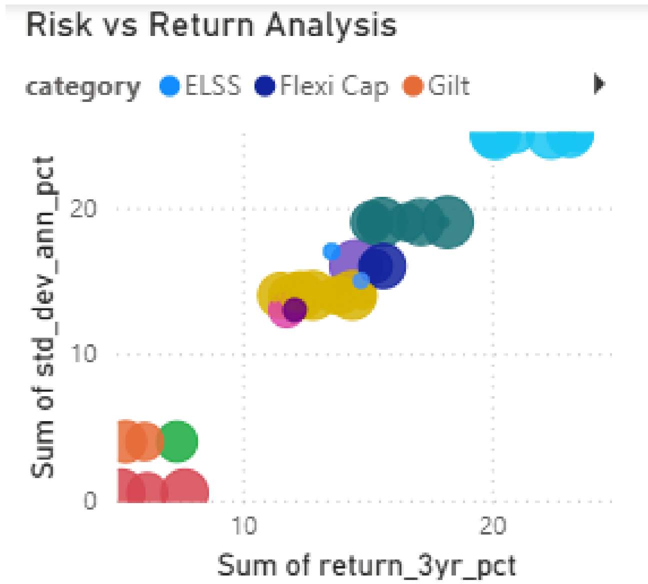


Figure 9: Risk vs Return Analysis

The scatter plot compares risk and return characteristics across mutual fund categories. Higher returns generally correspond to higher risk levels, while funds with superior risk-adjusted performance appear in the upper-right efficient region.

Observations

- High-return funds generally exhibited higher volatility.
- Some schemes delivered attractive returns with comparatively lower risk.
- Risk-adjusted analysis assisted in identifying efficient funds.

10. ROLLING SHARPE ANALYSIS

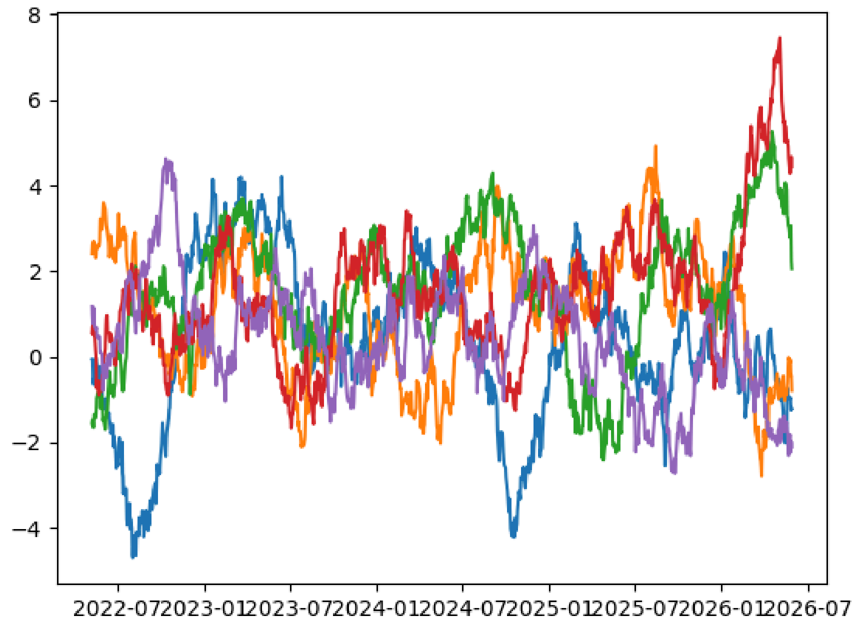


Figure 10: Rolling 90-Day Sharpe Ratio

Findings

- Sharpe Ratios varied across market cycles.
- Certain funds maintained stable risk-adjusted performance.
- Market volatility impacted short-term risk-adjusted returns.

11. INVESTOR ANALYTICS

Cohort Analysis

Investors were grouped by their first investment year.

SIP Continuity Analysis

Investors with transaction gaps greater than 35 days were identified as at-risk investors.

Findings

- Newer cohorts contributed significantly to investment growth.
- SIP discontinuity highlighted retention challenges.

12. PORTFOLIO CONCENTRATION ANALYSIS

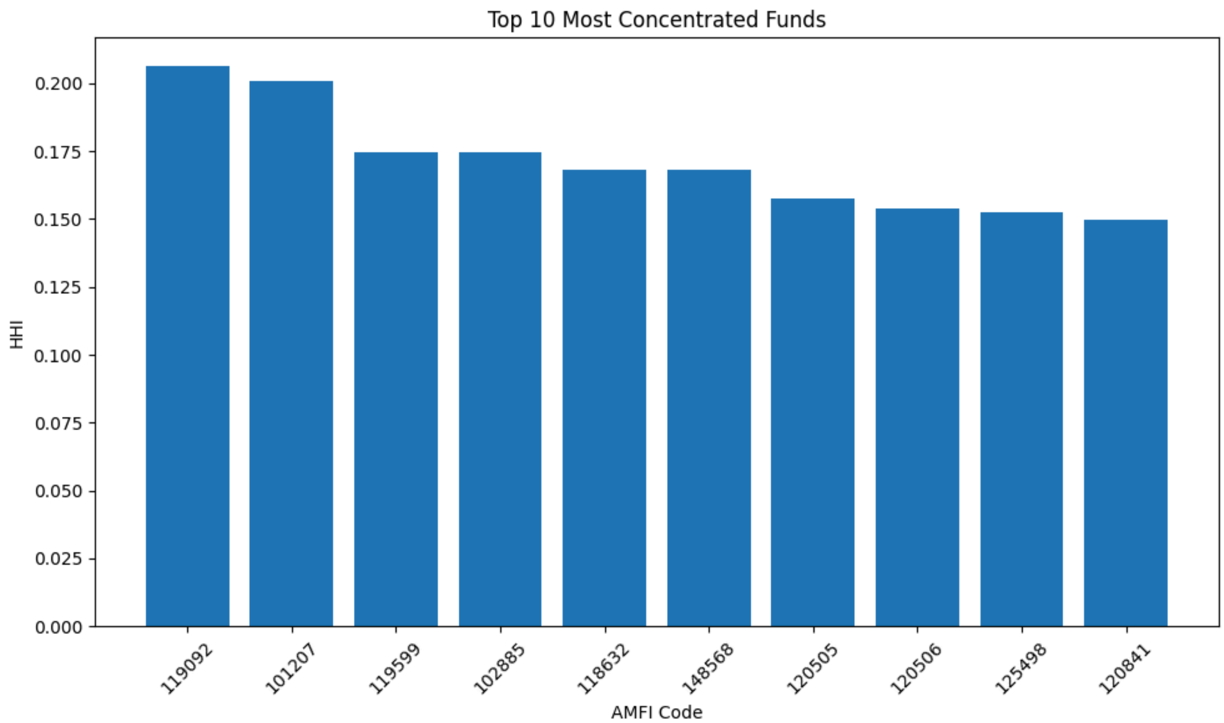


Figure 11: Sector Concentration Analysis (HHI)

Findings

- Diversified funds exhibited lower concentration risk.
- Certain schemes showed high sector dependency.
- HHI effectively quantified diversification levels.

13. DASHBOARD DEVELOPMENT

Industry Overview Dashboard

Provides a summary of industry growth, AUM, SIP inflows, folios, and leading fund houses.

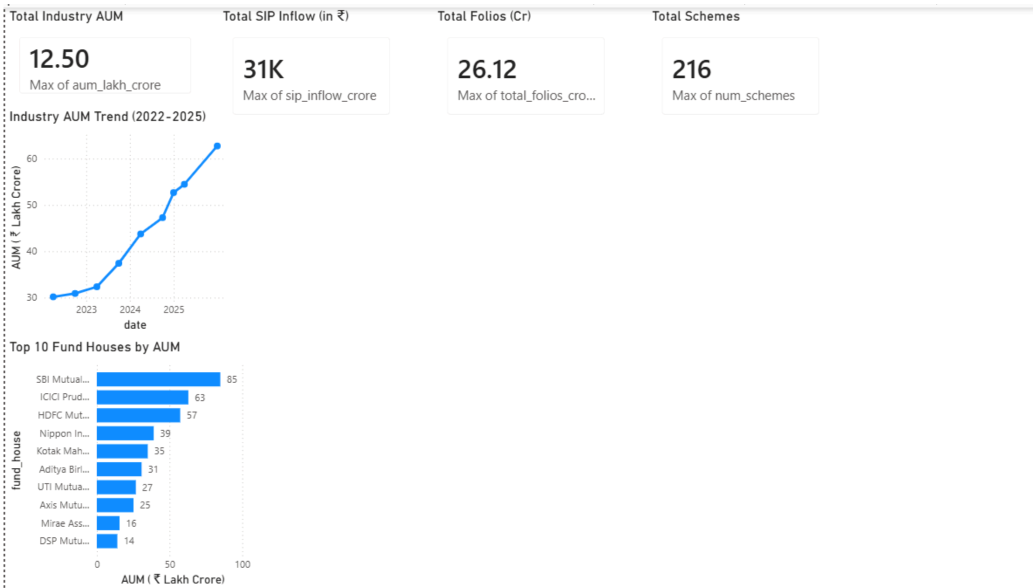
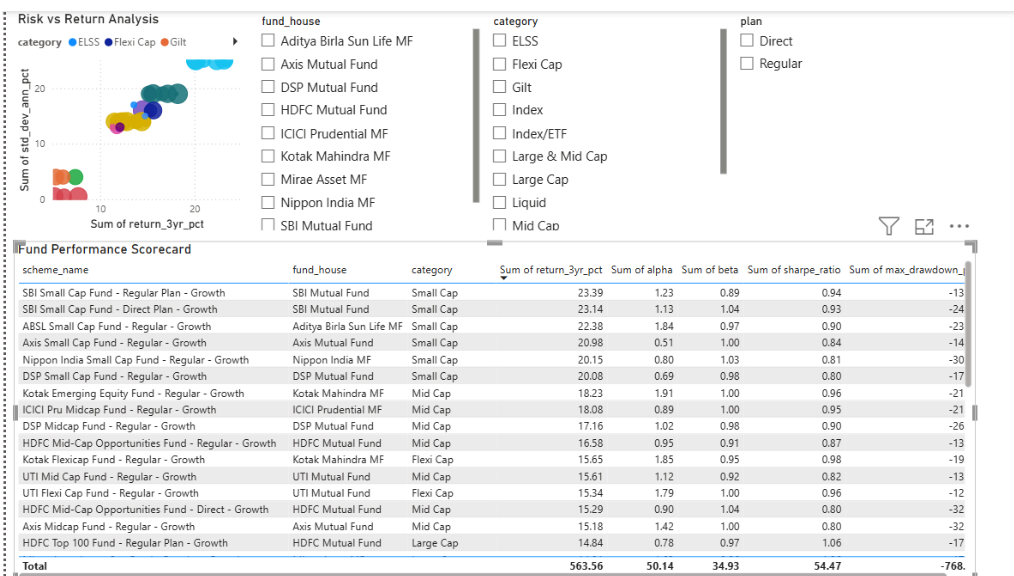


Figure 12: Industry Overview Dashboard

Fund Performance Dashboard

Analyzes risk-return characteristics and fund performance indicators.



scheme_name	fund_house	category	Sum of return_3yr_pct	Sum of alpha	Sum of beta	Sum of sharpe_ratio	Sum of max_drawdown_
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Small Cap	23.39	1.23	0.89	0.94	-13
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	Small Cap	23.14	1.13	1.04	0.93	-24
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	Small Cap	22.38	1.84	0.97	0.90	-23
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	Small Cap	20.98	0.51	1.00	0.84	-14
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	Small Cap	20.15	0.80	1.03	0.81	-30
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	Small Cap	20.08	0.69	0.98	0.80	-17
Kotak Emerging Equity Fund - Regular - Growth	Kotak Mahindra MF	Mid Cap	18.23	1.91	1.00	0.96	-21
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	Mid Cap	18.08	0.89	1.00	0.95	-21
DSP Midcap Fund - Regular - Growth	DSP Mutual Fund	Mid Cap	17.16	1.02	0.98	0.90	-26
HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	Mid Cap	16.58	0.95	0.91	0.87	-13
Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	Flexi Cap	15.65	1.85	0.95	0.98	-19
UTI Mid Cap Fund - Regular - Growth	UTI Mutual Fund	Mid Cap	15.61	1.12	0.92	0.82	-13
UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	Flexi Cap	15.34	1.79	1.00	0.96	-12
HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	Mid Cap	15.29	0.90	1.04	0.80	-32
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	Mid Cap	15.18	1.42	1.00	0.80	-32
HDFC Top 100 Fund - Regular Plan - Growth	HDFC Mutual Fund	Large Cap	14.84	0.78	0.97	1.06	-17
Total			**563.56**	**50.14**	**34.93**	**54.47**	**-768.**

Figure 13: Fund Performance Dashboard

Investor Analytics Dashboard

Analyzes investor demographics, transaction patterns, and investment behavior.

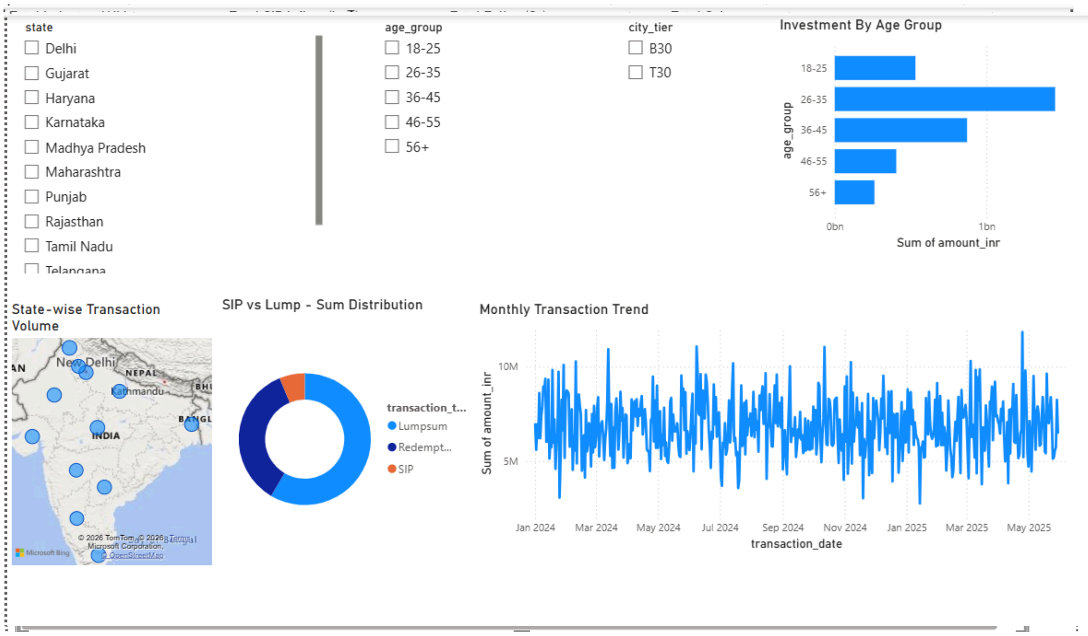


Figure 14: Investor Analytics Dashboard

SIP & Market Trends dashboard

Tracks SIP growth, category inflows, and benchmark market performance.

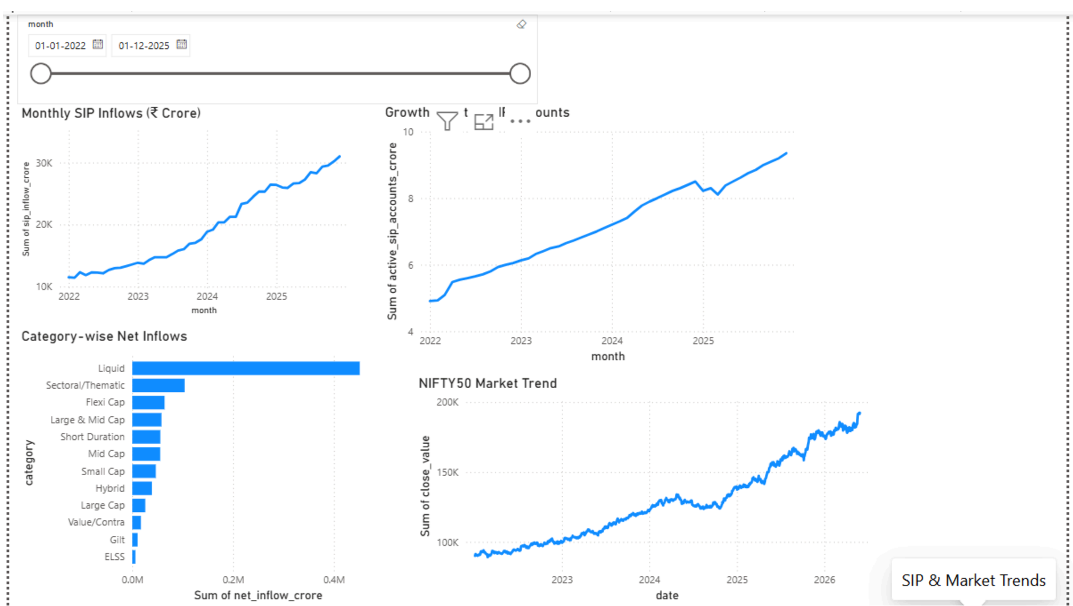


Figure 15: SIP & Market Trend Dashboard

14. KEY FINDINGS

- The mutual fund industry AUM showed consistent growth.
- SIP inflows reached record highs.
- Large-cap funds demonstrated stable performance.
- Risk-adjusted returns varied across categories.
- Investor participation increased significantly.
- Geographic distribution influenced investment behavior.
- Sector concentration varied across portfolios.
- Benchmark comparison identified outperforming schemes.

15. RECOMMENDATIONS

1. Investors should evaluate funds using risk-adjusted measures such as Sharpe Ratio rather than relying solely on returns.
2. Fund houses should closely monitor SIP discontinuity indicators to improve investor retention.
3. Diversification across sectors should be encouraged to reduce concentration risk.
4. Benchmark comparison metrics should be integrated into investor-facing platforms for improved transparency.
5. Automated dashboard reporting can significantly reduce manual reporting effort and improve decision-making.

16. LIMITATIONS

The Mutual Fund Analytics Platform was developed using available historical datasets and predefined business requirements. While the platform successfully provides valuable insights into mutual fund performance, investor behavior, and market trends, certain limitations exist.

1. **Historical Data Dependency**
   The analysis is based entirely on historical data. Future market performance and investment outcomes may differ significantly due to changing economic conditions and market dynamics.
2. **Limited Dataset Scope**
   The project utilizes selected mutual fund datasets provided during the internship. Additional factors such as macroeconomic indicators, interest rates, inflation data, and global market events were not incorporated into the analysis.
3. **Rule-Based Recommendation Engine**
   The fund recommendation system uses predefined risk categories and financial metrics rather than machine learning algorithms. Therefore, recommendations are based on static business rules and may not adapt dynamically to changing investor preferences.
4. **Simplified Risk Assumptions**
   Risk measures such as VaR, CVaR, Sharpe Ratio, and Beta are calculated using historical observations and assume that past market behavior can provide meaningful insights into future risk exposure.
5. **Dashboard Refresh Constraints**
   The Power BI dashboard currently operates on imported datasets and does not support real-time data streaming. Periodic updates are required to maintain current insights.
6. **Benchmark Coverage**
   Performance comparisons were conducted using selected benchmark indices. Inclusion of additional benchmarks could provide more comprehensive comparative analysis.

17. CONCLUSION

The Mutual Fund Analytics Platform successfully integrated data engineering, financial analytics, and business intelligence techniques into a unified analytical solution. The project demonstrated the complete lifecycle of data analytics, including extraction, transformation, storage, analysis, visualization, and reporting.

Using Python, SQL, SQLite, and Power BI, multiple datasets were transformed into actionable insights covering industry growth, fund performance, investor behavior, and portfolio risk. Advanced analytics including VaR, CVaR, Rolling Sharpe Ratio, Cohort Analysis, SIP Continuity Analysis, and Recommendation Systems further enhanced the analytical capabilities of the platform.

The final solution provides a scalable foundation for future mutual fund analytics applications and demonstrates the practical implementation of modern data analytics techniques within the financial services domain.

Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- Matplotlib
- Power BI
- Git
- GitHub
- Jupyter Notebook

References

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- BSE India
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